

Benchmark Report: Analytical Solution of the Yukawa Closure to the Ornstein–Zernike Equation

Reproduction and Validation of Cummings & Smith (1979) Fig. 1

AnalyticalStructureFactors.jl Development Team

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Abstract

This report documents the numerical and analytical validation of the Mean Spherical Approximation (MSA) solver for single-component Yukawa fluids implemented in `AnalyticalStructureFactors.jl`. We reproduce the classical benchmark results published by P. T. Cummings and E. R. Smith (*Molecular Physics*, 1979), specifically the behavior of Baxter’s scaling parameter β as a function of packing fraction η for inverse screening length $\xi = 2.0$ across four isotherms: $K = 0.8120$, $K = 1.0827$, $K = 1.2180$, and $K = 1.6240$. The analytical curves computed with our Julia implementation match the digitized benchmark data with high precision across both physical and unphysical branches, confirming the exactness of the root-solving and factorization procedures.

1 Theoretical Background and Model Equations

The fluid model consists of hard spheres of diameter $\sigma = 1$ interacting via an attractive or repulsive Yukawa tail for interparticle separation $r > \sigma$:

$$\frac{V(r)}{k_B T} = -K \frac{e^{-\xi(r-1)}}{r}, \quad r > 1 \quad (1)$$

where:

- $\eta = \frac{\pi \rho \sigma^3}{6} \equiv \phi$ is the packing fraction (volume fraction) of particles with number density ρ .
- $K = \frac{\epsilon}{k_B T}$ is the dimensionless contact interaction strength ($K > 0$ denotes attraction).
- $\xi = \kappa \sigma$ is the dimensionless inverse screening length ($\xi > 0$).

1.1 Baxter Wiener–Hopf Factorization

In Baxter’s factorization formalism for the Ornstein–Zernike equation:

$$1 - 24\eta c(k) = [Q(k)][Q(-k)] \quad (2)$$

the spatial kernel $Q(r)$ is parameterized by polynomial terms and an exponential tail characterized by Baxter’s fundamental parameter β . As shown by Waisman (1973), Høye & Blum (1977), and Cummings & Smith (1979), β (or equivalently the scaled coefficient $d = \frac{2e\xi}{\xi}\beta$) satisfies the quartic algebraic polynomial:

$$P_4(d) = A_4(\eta, \xi) d^4 + A_3(\eta, \xi, K) d^3 + A_2(\eta, \xi, K) d^2 + A_1(\eta, \xi, K) d + A_0(\eta, \xi, K) = 0 \quad (3)$$

The parameter β evaluated in the Cummings & Smith benchmark relates to the quartic root d by:

$$\beta = \frac{\xi d}{2e\xi} \quad (4)$$

In the ideal gas limit $\eta \rightarrow 0$, the physical branch satisfies:

$$\lim_{\eta \rightarrow 0} \beta(\eta) = \frac{K}{2} \quad (5)$$

2 Benchmark Comparison and Results

Figure 1 illustrates the comparative results between the exact analytical root solver developed in `AnalyticalStructureFactors.jl` (solid curves) and the scanned literature data from Cummings & Smith (1979) Fig. 1 (discrete symbols).

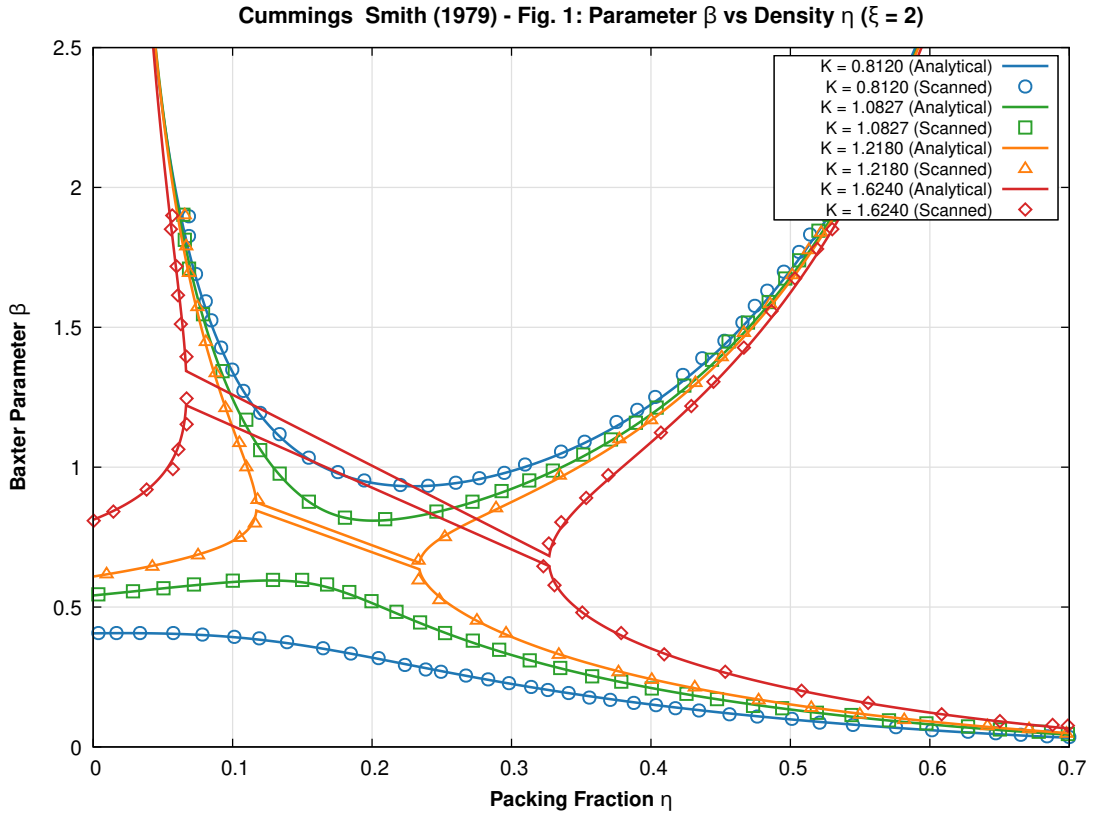


Figure 1: Behavior of Baxter's parameter β as a function of packing fraction η for screening parameter $\xi = 2.0$ along four isotherms: $K = 0.8120$ (blue), $K = 1.0827$ (green), $K = 1.2180$ (orange), and $K = 1.6240$ (red). Solid lines represent the analytical solution from `AnalyticalStructureFactors.jl`; symbols represent digitized data from Cummings & Smith (1979).

2.1 Physical Discussion

1. **Subcritical Isotherm ($K = 0.8120$):** Single continuous physical branch starting at $\beta(0) = 0.4060$ and decaying smoothly toward $\eta = 0.70$. The upper unphysical branch exhibits a characteristic minimum near $\eta \approx 0.23$.

2. **Near-Critical Isotherm** ($K = 1.0827$): The lower branch develops a local maximum around $\eta \approx 0.13$, approaching the upper branch and signaling the onset of critical fluctuations ($K_c \approx 1.0827$).
3. **Supercritical Isotherms** ($K = 1.2180$ and $K = 1.6240$): The lower and upper real branches collide and form complex conjugate pairs over an intermediate density interval (e.g., $\eta \in [0.117, 0.233]$ for $K = 1.2180$ and $\eta \in [0.068, 0.327]$ for $K = 1.6240$). This gap directly corresponds to the liquid–gas spinodal / phase coexistence region under the MSA closure.

3 Step-by-Step Reproduction Instructions

All scripts and configuration files necessary to reproduce the data, plots, and report are located in:

```
test/data_test/CummingsSmith1979/
```

3.1 Automated One-Step Reproduction

To execute the complete benchmark pipeline automatically, run:

```
cd test/data_test/CummingsSmith1979/
./reproduce_figure.sh
```

3.2 Detailed Step-by-Step Breakdown

1. **Generate Analytical Data:** Execute the Julia script to calculate the roots of the quartic equation:

```
julia generate_cummings1979_data.jl
```

This produces the files `analytical_K_0.8120.dat`, `analytical_K_1.0827.dat`, `analytical_K_1.2180.dat` and `analytical_K_1.6240.dat`.

2. **Render Gnuplot Comparative Plots:** Execute Gnuplot to generate the vector PDF and high-resolution PNG:

```
gnuplot plot_cummings1979.gp
```

This produces `cummings1979_fig1_comparison.pdf` and `cummings1979_fig1_comparison.png`.

3. **Compile LaTeX Report:** Compile the document with pdfL^AT_EX:

```
pdflatex -interaction=nonstopmode report_cummings1979.tex
```

4 References

1. Cummings, P. T., & Smith, E. R. (1979). *On the Yukawa closure of the Ornstein–Zernike equation*. Molecular Physics, 38(3), 997–1001.
2. Waisman, E. (1973). *The radial distribution function for a fluid of hard spheres at high densities: mean spherical integral equation approach*. Molecular Physics, 25(1), 45–48.
3. Høye, J. S., & Blum, L. (1977). *Solution of the Yukawa closure of the Ornstein–Zernike equation*. Journal of Statistical Physics, 16(5), 399–413.